```
from google.colab  import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
# importing
import pandas as pd
import plotly.express as px
from scipy import stats
import numpy as np
import statsmodels.api as sm
import statsmodels.formula.api as smf
import matplotlib.pyplot as plt
from math import sqrt

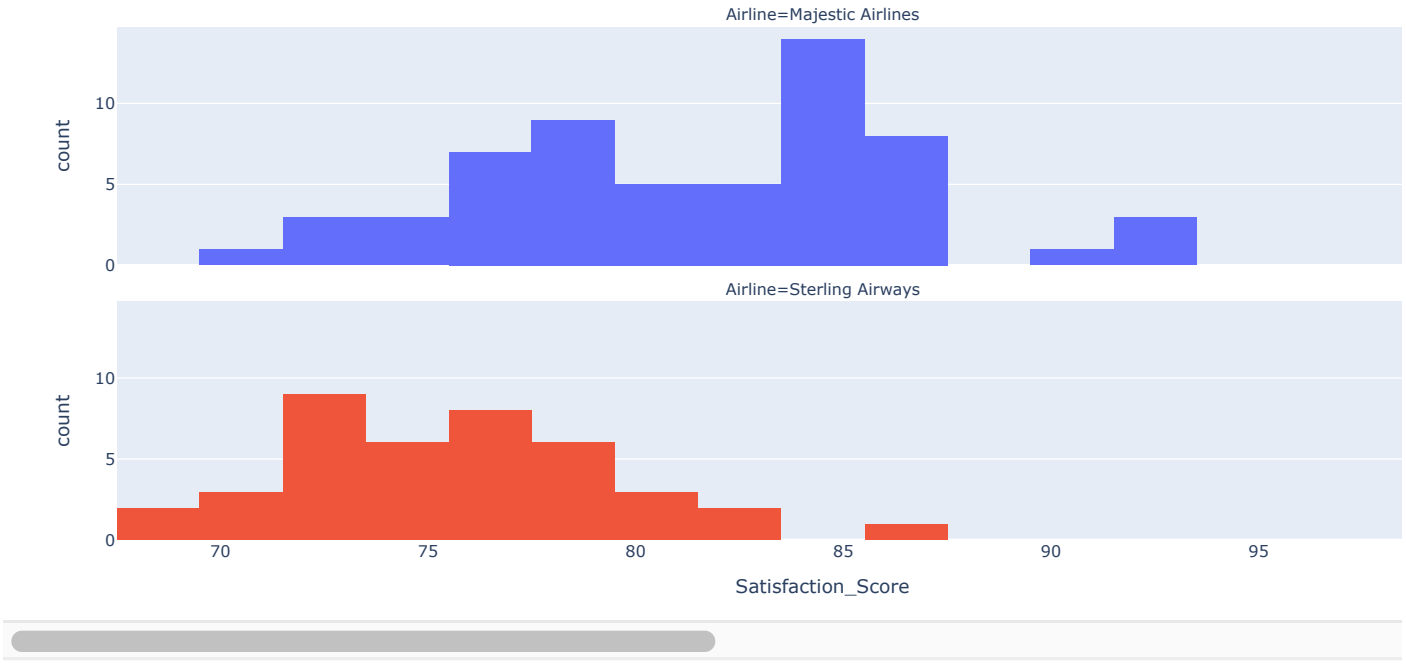
airlines = pd.read_csv('/content/drive/MyDrive/DSC325/airline_customer_satisfaction_survey.csv')
```

1. (a) The parameters of interest are the mean customer satisfaction score at Majestic Airlines and the mean customer satisfaction score at Sterling Airways

(b) Null hypothesis: The difference between the mean customer satisfaction score at Majestic Airlines and the mean customer satisfaction score at Sterling Airways is less than or equal to 4.

Alternative hypothesis: The difference between the mean customer satisfaction score at Majestic Airlines and the mean customer satisfaction score at Sterling Airways is greater than 4.

```
# (c) verifying conditions
# both samples are random and each is from a distinct population
# each sample measures the same variable
# means and standard deviations are unknown
# we may assume that the population is nearly Normally distributed
fig = px.histogram(airlines, x="Satisfaction_Score", facet_col="Airline",
                   facet_col_wrap=1, color="Airline")
fig.show()
```




```
# summary statistics
summary = airlines.groupby('Airline')['Satisfaction_Score'].describe()
print(summary)
```

```
                   count    mean       std   min   25%   50%   75%    max
Airline
Majestic Airlines   60.0  81.850  5.504467  71.0  78.0  82.5  85.0  100.0
Sterling Airways    40.0  75.575  3.908488  69.0  72.0  75.5  78.0   86.0
```

```
# sample size condition is satisfied because n > 10|K4|
n = len(airlines)
print("sample size =", n)
kurt = stats.kurtosis(airlines["Satisfaction_Score"])
print("10*abs(excess kurtosis) =", 10*np.abs(kurt))
```

```
sample size = 100
10*abs(excess kurtosis) = 5.271388936507915
```

```
# (d) test of significance
score = airlines["Satisfaction_Score"]
name = airlines["Airline"]
majestic = score[name == "Majestic Airlines"]
majestic2 = majestic - 4
sterling = score[name == "Sterling Airways"]

output = stats.ttest_ind(majestic2, sterling,
        alternative='greater',equal_var=False)
print(output)
```

```
TtestResult(statistic=2.4157156079670017, pvalue=0.008782670477585923, df=97.5658059339256)
```

(e) test statistic = 2.416

This was calculated using the formula: (mean1 - mean2)/squareroot((stdev1)^2/n1 + (stdev2)^2/n2)

This can be verified by the below calculation

```
mean1 = np.mean(majestic2)
mean2 = np.mean(sterling)
stdev1 = np.std(majestic2, ddof=1)
stdev2 = np.std(sterling, ddof=1)
n1 = len(majestic2)
n2 = len(sterling)
test_stat = (mean1 - mean2)/np.sqrt((stdev1**2/n1) + (stdev2**2/n2))
test_stat
```

```
2.4157156079670017
```

(f) P-value = 0.0088

Formula: P(T >= t)

degrees of freedom = 97.57

(g) Conclusion: Because the P-value < 0.05, we reject the null hypothesis at the 5% level

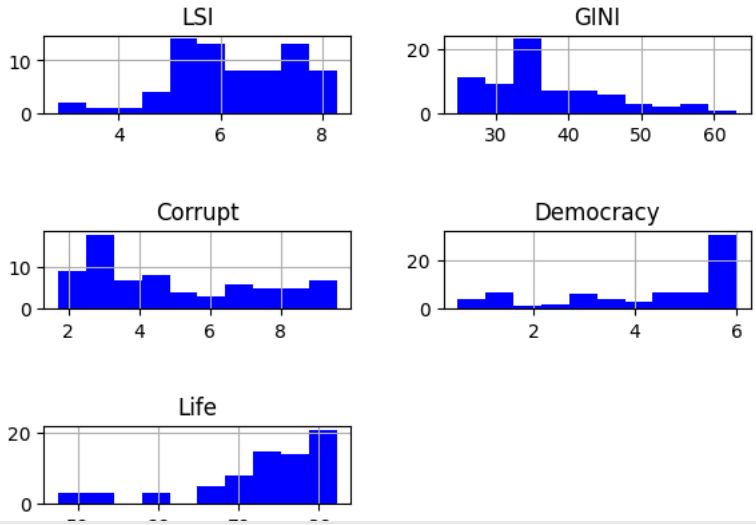
(h) Laymen's Terms: The data provides strong evidence that the mean customer satisfaction score of Majestic Airlines is greater than the mean customer satisfaction score of Sterling Airways by more than 4 points

```
happiness = pd.read_csv('/content/drive/MyDrive/DSC325/world_happiness_score.csv')
```

2. (a) First order model: y = b0 + b1x1 + b2x2 + b3x3 + b4x4 + e

We assume that errors, e, have a Normal probability distribution with mean equal to 0 and variance equal to σ^2. We also assume that random errors are independent, so the deterministic part of the model exists without e.

```
# (b) Histograms
happiness.hist(bins=10, color = "blue")
plt.tight_layout(pad=3.0)
plt.show()
```



GINI appears skewed to the right LSI, degree of democracy, and life expectancy appear skewed to the left

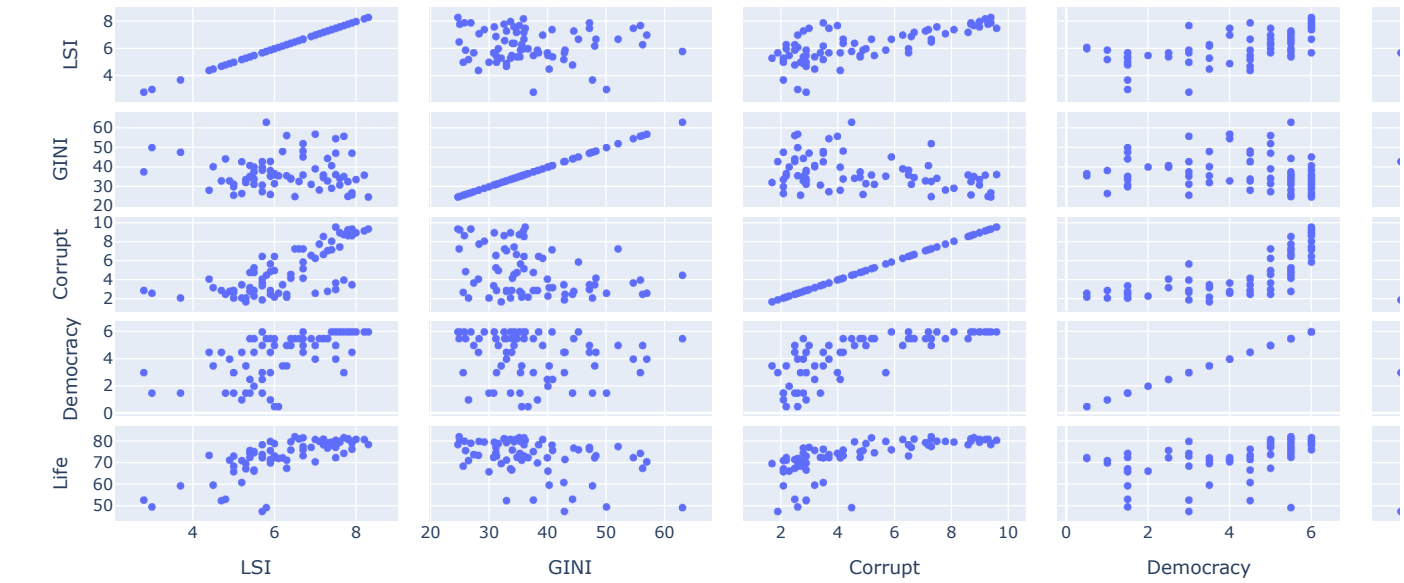
With this, life expectancy has some outliers on the lower range

Normality of the distribution of each independent variable is not a requirement for multiple linear regression

```
# (c) Descriptive Statistics
summary = happiness.describe()
summary
```

	LSI	GINI	Corrupt	Democracy	Life
count	72.000000	72.000000	72.000000	72.000000	72.00000
mean	6.238889	37.263889	4.909722	4.291667	72.97250
std	1.209418	8.579690	2.451396	1.679935	8.55398
min	2.800000	24.700000	1.700000	0.500000	47.56000
25%	5.475000	31.525000	2.800000	3.000000	71.05250
50%	6.150000	35.500000	4.150000	5.000000	74.67000
75%	7.300000	41.375000	7.125000	5.500000	79.55250

```
# (d) Scatterplots
scatters = px.scatter_matrix(happiness.iloc[:,1:7])
scatters.show()
```




```
quantitative = ["LSI","GINI","Corrupt","Democracy","Life"]
happiness2 = happiness[quantitative]
corr = happiness2.corr()
corr
```

	LSI	GINI	Corrupt	Democracy	Life
LSI	1.000000	-0.050248	0.697407	0.609227	0.721858
GINI	-0.050248	1.000000	-0.348463	-0.152189	-0.396138
Corrupt	0.697407	-0.348463	1.000000	0.747443	0.650336
Democracy	0.609227	-0.152189	0.747443	1.000000	0.524707
Life	0.721858	-0.396138	0.650336	0.524707	1.000000

It appears that degree of government corruption, degree of democracy, and life expectancy have a moderate relationship with LSI

```
# (e) multiple regression
happiness_df = sm.add_constant(happiness)
reg = smf.ols("LSI ~ GINI + Corrupt + Democracy + Life", data=happiness_df)
res = reg.fit()
res.summary()
```

OLS Regression Results

Dep. Variable:	LSI	**R-squared:**	0.702
Model:	OLS	**Adj. R-squared:**	0.685
Method:	Least Squares	**F-statistic:**	39.53
Date:	Fri, 22 Nov 2024	**Prob (F-statistic):**	5.72e-17
Time:	21:30:23	**Log-Likelihood:**	-71.720

```
anova_table = sm.stats.anova_lm(res, typ=2)
anova_table
```

	sum_sq	df	F	PR(>F)
GINI	8.316863	1.0	18.028594	6.858726e-05
Corrupt	5.332832	1.0	11.560063	1.139340e-03
Democracy	0.233780	1.0	0.506768	4.790128e-01
Life	17.039586	1.0	36.936978	6.499455e-08

Skew:	-0.187	**Prob(JB):**	0.788

(f) Model of utility

Null Hypothesis: All "b" coefficients are equal to each other, equal to 0

Alternative Hypothesis: At least one "b" coefficient is not equal to 0

F-statistic = 39.53

(Formula for F-statistic: (mean square of model)/(mean square of error))

P-value = 5.72e-17

(Formula for P-value: P(F>Fc))

Conclusion: The P-value is very close to 0, indicating that the model is statistically useful, in that at least one explanatory variable may contribute to the prediction of LSI

(g) Least squares prediction equation:

LSI = (0.0447)x1 + (0.1941)x2 + (0.0526)x3 + (0.0782)x4

```
# (h) RSE
SSE = res.ssr
MSE = SSE/(len(happiness) -2)
RSE = sqrt(MSE)
RSE
```

```
0.6644881246529222
```

Interpretation: most (95%) observed y-values lie within 2s, or 1.33, of the least squares line

(i) R^2 = 0.702

Interpretation: About 70.2% of the variation in y can be attributed to using x to predict y in this model

Rα2 = 0.685

Interpretation: Using the 4 independent variables in this first order model explains 68.5% of the total sample variation in LSI when sample size and number of b parameters are considered

(j) Significance of democracy variable

Null hypothesis: the coefficient for degree of democracy is equal to 0

Alternative hypothesis: the coefficient for degree of democracy is not equal to 0

t-statistic = 0.712

P-value = 0.479

Conclusion: Because the P-value is quite large, we fail to reject the null hypothesis, and degree of democracy is not indicated to be a useful predictor of LSI

95% confidence interval = [-0.095, 0.200]